

Business Requirements Document (BRD)

1. Document Control

- System Name: Application

2. Executive Summary

- This document outlines the business requirements for the Application.

3. Business Objectives

- Ensure data from source CSV files is loaded efficiently into delta tables.
- Maintain data quality by removing nulls and duplicates.
- Enable historical data tracking via SCD Type 2.
- Provide aggregated spend insights per customer per date.
- Automate ETL processes using delta live tables.

4. In-Scope

- Read and load customer and order data into delta tables.
- Implement schema as specified for customer and order tables.
- Add calculated fields (e.g., TotalAmount).
- Data cleansing processes.
- Creation and management of ordersummary and customeraggregatespend tables.
- SCD Type 2 implementation for ordersummary.
- Aggregation logic for customer spending.
- Execution using delta live tables.

5. Out-of-Scope

- Source CSV data creation or modification.
- Upstream data capture or external ETL tools.
- Downstream analytics or BI reporting beyond specified tables.
- Integration with systems not mentioned in requirements.

6. Stakeholders

- Data Engineering Team
- Business Analysts
- Data Governance Team
- Application Owners
- End-Users (Business Consumers)

7. Business Requirements

BR-BRD-001 — Source Data Loading

- Description: Read source CSV data from specified volume and load into delta tables (customer, order).
- Business Value: Centralizes and structures raw data for further processing.
- Priority: High

BR-BRD-002 — Schema Application

- Description: Apply specified schema to customer and order tables.
- Business Value: Ensures data consistency and usability.
- Priority: High

BR-BRD-003 — Calculated Fields Addition

- Description: Add "TotalAmount" column to order table as multiplication of PricePerUnit and Qty.
- Business Value: Enables financial analytics on order data.
- Priority: High

BR-BRD-004 — Data Cleansing

- Description: Remove null and duplicate records from both tables.
- Business Value: Improves data quality and reliability.
- Priority: High

BR-BRD-005 — Ordersummary Table Creation

- Description: Create ordersummary table with defined schema in specific catalog and schema.
- Business Value: Facilitates consolidated order data for downstream processes.
- Priority: High

BR-BRD-006 — SCD Type 2 History Maintenance

- Description: Implement SCD Type 2 logic to maintain history; update ordersummary when customer data changes.
- Business Value: Enables change tracking and historical analysis.
- Priority: High

BR-BRD-007 — Customer Spend Aggregation

- Description: Create customeraggregatespend table and load aggregated "TotalAmount" per customer per date.

- Business Value: Provides customer spend insights for business stakeholders.
- Priority: High

BR-BRD-008 — Automated Processing with Delta Live Tables

- Description: Execute all ETL steps using delta live tables framework.
- Business Value: Enables automation, reduces manual effort, and increases reliability.
- Priority: High

8. Assumptions & Constraints

- Source CSV files are available and accessible at specified paths.
- Data conforms to given schema definitions.
- Delta live tables environment is properly configured.
- System resources and privileges are sufficient for job execution.
- External integrations are out of scope.

9. Success Metrics

- Successful ingestion of all records from source files.
- Zero null or duplicate records in final tables.
- Accurate calculation and aggregation of TotalAmount.
- SCD Type 2 logic correctly maintains historical versions.
- Automated runs complete within target SLAs.

10. Risks & Mitigations

- Source data unavailable or incorrectly formatted — Mitigation: Exception handling and monitoring.
- ETL failures due to environment issues — Mitigation: Pre-run validation, error logging.
- SCD Type 2 logic defects — Mitigation: Review and rigorous testing.
- Data volume growth impacting performance — Mitigation: Monitor and scale as needed.

11. Approval